INCH POUND

MIL-C-3098/96D
5 May 1993
SUPERSEDING
MIL-C-3098/96C
8 March 1983

MILITARY SPECIFICATION SHEET

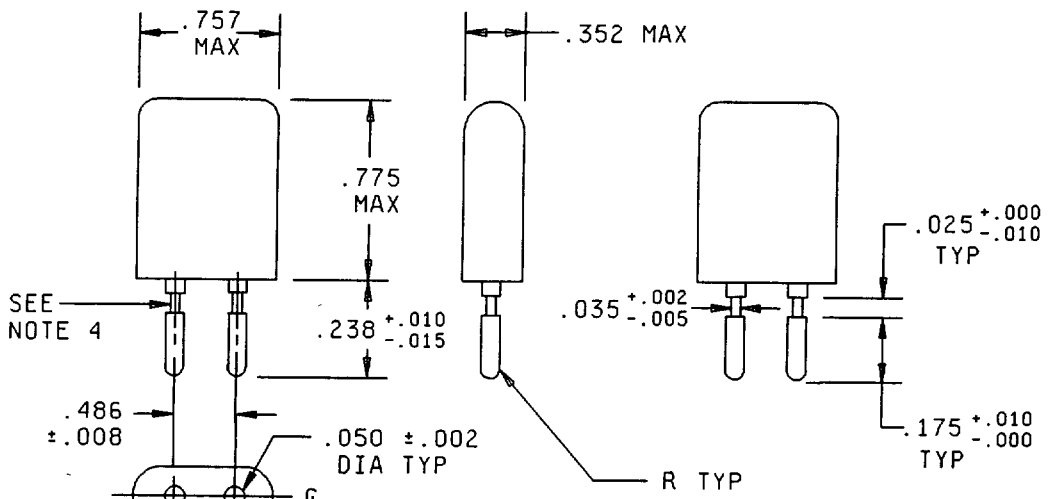
CRYSTAL UNIT, QUARTZ, CR90/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

INACTIVE FOR NEW DESIGN

Pertinent characteristics: 3 to 5 MHz; fundamental; controlled; antiresonance.



Inches	mm	Inches	mm
.002	.05	.050	1.27
.005	.13	.175	4.45
.008	.20	.238	6.05
.010	.25	.352	8.94
.015	.38	.486	12.34
.025	.64	.757	19.23
.035	.89	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR90/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-27/U (MIL-H-10056/14).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 3 to 5 MHz, inclusive.

Frequency tolerance: ±4 ppm at +30°C ±0.5°C.

Equivalent resistance: 3 MHz 15 ohms, maximum.
5 MHz 9 ohms, maximum.

Antiresonance load capacitance: 30 ±0.5 pF.

Mode of oscillation: Fundamental.

Operating temperature range: -55°C to +90°C, inclusive.

Capacity ratio $\frac{C_0}{C_1}$: 310 ± 10.

Drive level, rated: 10 µW.

Frequency deviation between -40°C ±0.5°C and upper turning point: Within ± 5 ppm.

Crystal quality (Q): $Q \geq .5 \times 10^6$ at +30°C. 1/

Spurious coupling: The rate of change of crystal resistance with temperature shall not exceed 1 ohm per 3°C over the entire temperature range of -55°C to +90°C.

Capacitance, shunt: 3 ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging: 2/

Temperature: 65°C ±1°C. The instability of temperature control during the aging test shall not exceed ±0.01°C. The average weekly aging rate shall be no greater than .01 ppm per week for the last 2 weeks of a 4-week aging test.

1/ $Q = \frac{1}{4\pi(\Delta f)(C_T)(R)}$

Where: Δf = difference, in Hz, between the series and antiresonant frequency at C_L (load capacitance) of 30 pF.
C_T = Capacitance: C_o (shunt capacitance of the crystal unit) plus C_L (load capacitance) of 30 pF.
R = Motional-arm resistance of the crystal unit.

2/ The crystal units shall be preconditioned by storage in an oven at +65°C ±5°C for 3 weeks at the time of manufacture.

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Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent:

QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE: NATO guidance type NXT 45.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:

Army - ER
Navy - EC
Air Force - 85

Review activities:

Army - AR, MI, SM

User activities:

Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:

Army - ER

Agent:

DLA - ES

(Project 5955-0678-11)

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